

Adam Zhang

☎ (636) 706-2249 | ✉ reachadamzhang@gmail.com | 🔗 LinkedIn | 🐙 GitHub | 🌐 adamzhang.org

EDUCATION

Purdue University

West Lafayette, IN

Bachelor of Science in Artificial Intelligence | GPA: 3.93

Expected May 2028

- John Martinson Honors College (JMHC) – Fall 2026 Cohort
- Study Abroad program across Japan (May & Jun. 2026), expanding global perspective and cultural adaptability.

PROJECTS

StyleSnapped | PyTorch, Python, FastAPI, React, sklearn, Gemini API

Apr. 2026

- Presented an end-to-end AI web application at the ML@Purdue Symposium that classifies clothing aesthetics using a custom fine-tuned **ResNet-50** vision model served via **FastAPI** and **React**.
- Engineered a semi-supervised data pipeline to label **44,000** images by extracting 2048-dimensional embeddings, clustering with **K-Means**, and propagating **Gemini VLM** centroid labels to bypass API costs.
- Achieved **83% validation accuracy** despite a 14x class imbalance by implementing a two-phase progressive unfreezing training strategy and applying inverse-frequency class weights in **PyTorch**.

AXI6 Cinema Robotics | Python, JavaScript, React, FastAPI, OpenCV, Raspberry Pi

Feb. 2026 - Apr. 2026

- Engineered a 2-axis robotic camera slider for the StarkHacks hackathon, reducing commercial equipment costs to **\$200** by integrating a **Raspberry Pi** and custom 3D-printed mechanics.
- Enabled timeline-based waypoint animation and remote manual control by developing a custom web interface utilizing a **React** frontend and **FastAPI** backend.
- Achieved smooth ramping during lateral movement by implementing an axis synchronization algorithm in **Python** that dynamically computes and executes counter-rotations.

BarBabes | Python, JavaScript, React Native, FastAPI, MongoDB

Feb. 2026

- Won **1st Place Overall** (of 33 teams) and **Best Use of AI** at Innovate HER Hacks 2026 by engineering a holistic impairment assessment platform.
- Managed real-time data and seamless **NFC “tap-to-log”** events by building a cross-platform app with **React Native**, **FastAPI**, and **MongoDB**.
- Assessed user impairment holistically by integrating the **Google Gemini API** to analyze BAC results, reaction latency, and drinking history.

EXPERIENCE

CS Undergraduate Student Board

Mar. 2026 – Present

Board Member

- Amplified student engagement by directing **digital media strategies** to increase department visibility.
- Advocated for the undergraduate body by collaborating with **Purdue CS department heads** and faculty.

AIDA3 Research - AI for Autonomous Aviation

Jan. 2026 – Present

Undergraduate Researcher in VIP

- Earned the **Outstanding First-Year Student Award** (Spring 2026) for exceptional technical contributions, dedication, and demonstrated initiative within the Vertically Integrated Projects (VIP) program.
- Engineered path planning for UAVs by applying **deep reinforcement learning** to GNC.
- Simulated fixed-wing flight dynamics by creating custom **OpenAI Gym** environments tailored for a **DRL agent**.
- Validated agent performance across diverse conditions by developing a cross-validation wrapper in **ROS2**.

St. Louis University

May 2023 – Aug. 2023

Software Development Intern

- Accelerated RNA sequencing analysis by **36.6%** with **R** to optimize Transcription Start Site (TSS) processing.
- Doubled **BAM file** read speeds by implementing **multicore processing** pipelines for Windows and Linux.

FIRST Tech Challenge - Robotics

Sept. 2022 – Apr. 2025

Lead Programmer and AI Camera Vision

- Ranked **85th globally** (of ~8,000 teams) and won the **2025 MO State Championship**.
- Optimized robot motion and target detection by fine-tuning **PID motor control** and camera vision in **Java**.
- Improved autonomous navigation by training custom **machine learning** vision models with **TensorFlow**.

TECHNICAL SKILLS

Languages: Python, Java, R, C++, JavaScript, HTML/CSS, SQL, NoSQL

Frameworks: React, ROS 2, React Native, FastAPI, MongoDB, Tailwind CSS

Libraries: Pandas, NumPy, sklearn, PyTorch, TensorFlow, Keras, OpenCV, Matplotlib, Gymnasium, Seurat

Tools: Git, GitHub, Claude Code, VS Code, Antigravity, RStudio, PyCharm, Android Studio